

Crossed effects of audio-visual environment on indoor soundscape perception for pleasant open-plan office environments

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ABSTRACT

An open-plan office (OPO) layout promotes communication and teamwork among workers and provides flexibility in working; however, it also adversely affects work satisfaction, health, and worker privacy, leading to dissatisfaction and poor task performance. In this study, the subjective responses in an audio-visual OPO environment were analyzed to investigate the effects of the environment on indoor soundscape perception for designing pleasant OPOs. The outdoor environments of four OPOs with four different visual layouts were divided based on two conditions with different window views (good and bad views) to model eight visual stimuli. The auditory aspect was evaluated by computer simulation using three sound sources. For subjective assessment, virtual reality technology was employed with 24 audio-visual stimuli, and thirty-four subjects were asked to rate the perceived affective quality and overall quality. Task performance was quantified using auditory digit span tasks. The spatial preference decreased with the Loud, Variable, and Reverberant attributes and increased with the Bright, Orderly, and Wide attributes. For the same OPO soundscape, the window area ratio and floor area showed positive correlations with visual privacy, visual satisfaction, speech privacy, and sound satisfaction. However, work satisfaction and task performance exhibited contradictory relationships with the visual parameters. Furthermore, wide, open workspaces yielded high work satisfaction, whereas narrow, confined work spaces yielded high task performance. These results indicate that the relationship between work satisfaction and task performance must be considered for a balanced OPO design, providing practical guidance for designing pleasant OPO environments.

1. Introduction

1.1. Importance of sound environment in an open-plan office

An open-plan office (OPO) is a widely used layout in offices given that it promotes communication and teamwork among workers and provides high space flexibility at low costs [1,2]. However, some studies have reported that an OPO layout significantly influences work satisfaction, health, and privacy of the workers [3,4]. Zhang et al. [5] conducted a social survey on the general working environment of an OPO and found that the workers were highly dissatisfied with the privacy and environmental acoustics. Kim and de Dear [6] conducted a general survey and stated that among various factors that determine workplace satisfaction in an OPO such as thermal, air, lighting, acoustics, layout, furnishings, and cleanliness, the most essential were speech privacy, visual privacy, and noise level. Therefore, based on the results of previous studies, we can conclude that noise, auditory distraction, and lack

of privacy are key factors that cause dissatisfaction in OPOs [6–12]. Particularly, high noise levels increase worker fatigue and reduce cognitive skills such as memory, thereby having a negative effect on work satisfaction and task performance in OPOs [13–16]. Among the various types of noise in OPOs, conversational noise and ringing of telephones were reported as the main causes for the deteriorating concentration of workers [13]. Several studies stated that irrelevant speech or intelligible conversation among fellow workers deteriorates the efficiency and performance of workers in OPOs and increases workload [17–19]. In addition, noise due to low-frequency ventilation results in reduced social orientation and discomfort [20].

Because a poor acoustic environment in OPOs can deteriorate speech intelligibility and result in reduced work performance [21], worker privacy is essential to improve work satisfaction and task performance [22]. Privacy in OPOs can be classified into speech and visual privacy relating to sound and vision, respectively. Speech privacy refers to the ability to control the environment such that private conversations are

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unheard outside the workspace, whereas visual privacy refers to the ability to prevent outsiders from seeing workspace occupants [23]. Several attempts have been made using architectural elements and interior design techniques, such as workstations, partitions, and office layouts, to ensure the privacy of office spaces [16,24]. While most studies focused on noise and provided ways to secure speech privacy, very few have considered visual privacy [9,12,25,26]. Goins et al. [25] examined the relationship between speech and visual privacy and the architectural enclosure of an OPO but was unable to determine the relationship between visual privacy and work performance owing to limited data. However, no studies have comprehensively considered both sound and visual privacy of OPOs or clearly determined their relationship with work satisfaction and task performance.

1.2. Soundscape concept for an indoor environment

Several studies have been conducted to apply the emerging concept of soundscape in outdoor environments such as urban areas, rural areas, parks, and open spaces for design purposes [27,28]. Soundscape refers to the acoustic environment as perceived, experienced, and/or understood by humans. Considering that surrounding context is essential in the interpretation of soundscape [29], various attempts have been made to consider non-acoustic contexts such as visual, daylight, odor, socio-cultural, and individual, in addition to acoustic factors [30]. Consequently, relevant predictive models were developed in a study to evaluate soundscapes in outdoor environments [31]. Additionally, in recent studies, the concept of soundscape in indoor environments was applied [32–37]. Dokmeci and Kang [36] proposed a questionnaire draft to assess indoor soundscapes from different perspectives. Torresin et al. [35] suggested the use of comfort, content, and familiarity as cognitive scales to interpret indoor soundscapes. Unlike outdoor soundscapes, space function, physical conditions, and spatial characteristics should be considered for interpreting indoor soundscapes [36]. While previous studies have assessed soundscapes in indoor environments such as residential buildings, libraries, historical spaces, and public spaces based on the functional aspect of space, no studies have assessed OPOs [32–37]. Therefore, in this study, we assess indoor soundscapes in OPOs, where workers spend most of their day.

1.3. Multi-sensory approach for the perception of indoor soundscapes

Many studies have analyzed audio-visual interactions among various senses, given that a multi-sensory approach is vital for the holistic interpretation of soundscapes [38–41]. To evaluate soundscapes using a multi-sensory approach, the interaction (combined effect), which examines one domain (e.g., satisfaction) using two or more other domains (e.g., visual and auditory stimuli), and the crossed effect which examines the significant effect of one domain (e.g., visual stimuli) on another domain (e.g., acoustic satisfaction), must be defined distinctly [42,43]. Jeon and Jo [41] suggested that the perception of soundscape and landscape varied depending on audio-visual interactions, which is a key factor for determining satisfaction with space. Furthermore, the perception of a space can be changed by the people present in the space based on the sounds they create [44]. Therefore, it is important to comprehensively analyze the effect of audio-visual interaction and crossed effect, particularly in spaces where a single space is shared by multiple people, such as an OPO. Based on this, several studies have been conducted on audio-visual interactions for soundscape design in indoor environments [45]. In addition, the influence of audio-visual information on the perception of residential noise (floor impact noise, traffic noise, etc.) has also been examined in studies [46,47]. Abdalrahman and Galbrun [48] conducted a study in an OPO to analyze the changes in the preference for various water features based on audio and visual interactions and found that water sounds at 45 dBA were the most significant for an office environment, where 48 dBA is the sound level at which irrelevant speech typically occurs. Furthermore, preference

improvement effects of up to 2.5 times were observed when a visual water feature was installed at an appropriate location compared to when only the sound was provided, which enabled the use of water sounds to mask other sounds in the office design [48]. Nevertheless, research on audio-visual interaction and crossed effects in OPOs remains limited.

1.4. Utility of virtual reality technology for indoor soundscape assessment

OPOs can be assessed through observational and interventional studies [45]. Observational studies survey large populations in office environments using a questionnaire to derive generalized conclusions, whereas interventional studies derive more controlled conclusions by providing the same stimuli to subjects under laboratory conditions [49]. Although laboratory studies directly reproduce or rent spaces similar to an office and provide only sound information using a separate playing device [16,50,51], reproducing a space incurs high costs, and sound information alone may not be ecologically valid. These limitations are overcome using virtual reality (VR) technology [46,52], considering it in analyzing the human perception of various indoor noises has already been proved [47,53] in previous studies that used VR technology to assess soundscapes in OPOs and propose specific designs. For example, when the window-to-wall ratio in an office is greater than 30% in VR, compared to the usual 15%, worker satisfaction increases by approximately 1.9 times, enabling the utilization of window designs in offices [53]. Nevertheless, there are limitations to VR technology. First, there may be differences in the senses of immersion and realism with respect to the soundscape when using a non-individual head-related transfer function (HRTF). This is because the HRTF implementing the directionality of sound varies from person to person [47]. Furthermore, soundscape perception may vary depending on whether head-tracking technology is applied [54]. Moreover, VR technology may lead to side effects such as cybersickness and discomfort or give rise to limitations in behavioral responses such as the Proteus effect or dissociative disorders in virtual space, which differ from those of real spaces [55]. Despite these limitations, VR technology is useful for evaluating studies within the laboratory environment, and hence, we implement a laboratory assessment environment using VR technology in this study [48].

1.5. Study objectives

As mentioned above, even though numerous scholars have evaluated soundscapes in outdoor spaces, there is a lack of research on soundscapes in indoor spaces, especially in OPOs, where the soundscape is considered essential based on the characteristics of spatial functions. In addition, there has been a lack of research on improving overall working environment quality considering multi-sensory aspects, such as studies of work and task performance in an office. Therefore, in this research, the audio-visual crossed effects within an office environment were investigated, the relationship between work satisfaction and task performance was examined, and an indoor soundscape design strategy for a pleasant OPO layout was developed. The main research objectives of this study were as follows:

- 1) To investigate the crossed effects of audio-visual physical parameters on the perceived affective quality of an OPO environment.
- 2) To examine the crossed effects of audio-visual physical parameters on the overall quality of an OPO environment.
- 3) To determine the crossed effects of audio-visual attributes on the overall quality of an OPO environment and propose a strategy to improve work satisfaction and productivity.

2. Methods

2.1. Study design

As shown in Fig. 1, this study uses a conceptual framework to create a

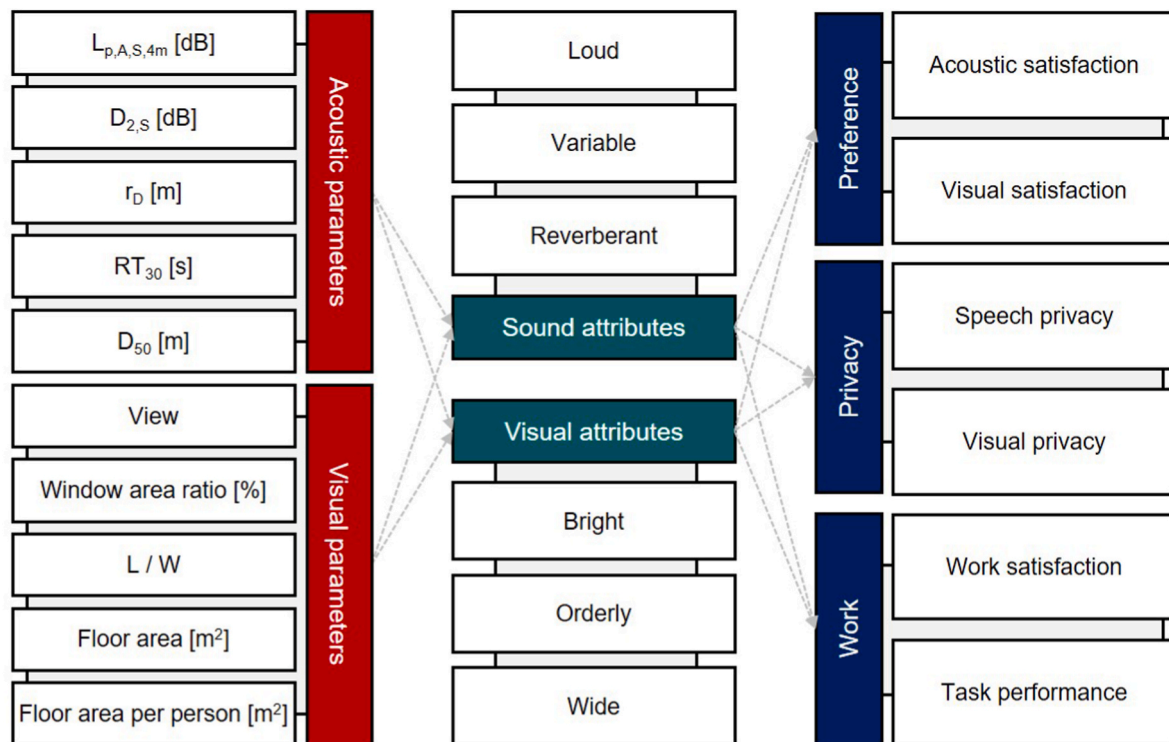


Fig. 1. Conceptual framework for this study.

pleasant indoor soundscape in an OPO layout. Audio stimuli were generated through computer simulations, whereas visual stimuli were created by developing virtual office spaces with different layouts using VR technology to set up an assessment environment similar to the actual settings within the laboratory. The survey was conducted on the same group of subjects using a within-subjects design [56], which examined the relationship between the overall survey-based quality and the physical properties of the audio-visual environment in the office and interpreted the results based on the audio-visual crossed effect for

application in soundscape design for OPOs.

2.2. Audio stimuli

2.2.1. Field measurement

As shown in Fig. 2(a), the information necessary for the computer simulation to generate audio stimuli was collected by conducting on-site measurements in a typical OPO space having dimensions of 20.2 m (length) $\times$ 15.2 m (width) $\times$ 2.5 m (height), based on ISO 3382-3 [57]

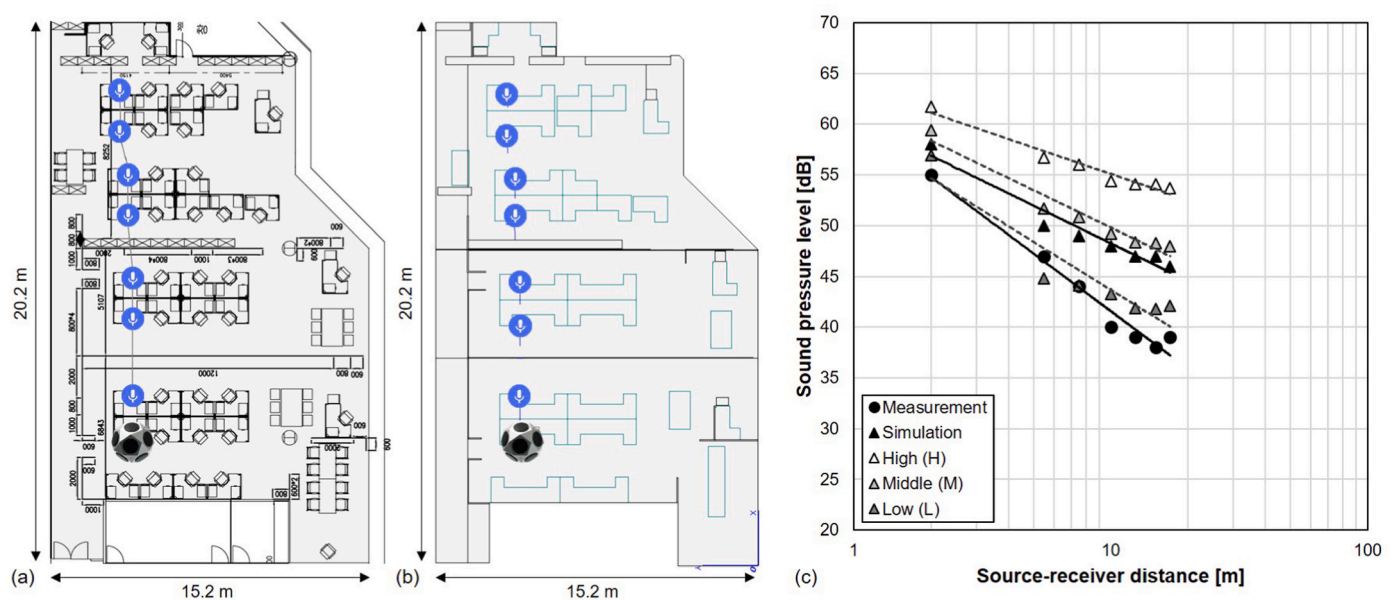


Fig. 2. Comparison of acoustic measurements and computer simulation. (a) Measurement setup (blue circles indicate the measurement positions), (b) Simulation setup, and (c) Sound pressure level based on the source-to-receiver distance. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

and ISO 3382-1 [58]. The ceilings and floors of the OPO were finished with acoustical boards (absorption gypsum boards) and vinyl tiles, with 1.1 m partitions installed inside the OPO. We used an omnidirectional speaker (Type 4292, Brüel & Kjær) and a 1/2-inch microphone (GRAS AE 46, GRAS Sound and Vibration) to record the measurements. Furthermore, we employed sine sweep and omnidirectional sound sources, which met the sound power spectrum of general voices presented in ANSI S 3.5-1997 (R 2007) and selected seven measurement points depending on the line crossing the unit workspace.

Three single-number quantities— $L_{p,A,S,4m}$, the A-weighted sound pressure level of speech at 4 m from the sound source; $D_{2,S}$, the spatial decay rate of speech; and r_D , the distraction distance—presented in ISO 3382-3, and two acoustic parameters— RT_{30} , the reverberation time, and D_{50} , the definition—presented in ISO 3382-1, were selected to analyze the recorded sound sources, as summarized in Table 1 [12,59]. $D_{2,S}$ refers to the decibel decrease in the A-weighted sound pressure level as the distance doubles, calculated from the linear regression within 2–16 m, r_D is the distance at which the speech transmission index from the speaker decreases below 0.5, RT_{30} is determined by doubling the time interval between the initial 5 dB decrease and the 35 dB decrease in the energy decrease curve after the sound source stops playing, and D_{50} is the ratio of the initial 50 ms energy to the total sound energy. Consequently, the values were determined as: $L_{p,A,S,4m} = 49.4$ dB, $D_{2,S} = 4.6$ dB, and $r_D = 11.1$ m. RT_{30} and D_{50} were calculated as 0.6 s and 0.89 s, respectively.

2.2.2. Acoustic 3D modeling

As shown in Fig. 2, a computer simulation model similar to an actual office space was created, representing the general working environment in Korea, based on the results of on-site measurements performed to develop the characteristics of the actual soundscape of a general office environment through computer simulation [Fig. 2(b)]. We used the ODEON simulation version 15 for the computer simulation. The ceilings, walls, floors, partitions, desks, and drawers of the actual office were modeled in 3D for the indoor design factors, and sound absorption coefficients similar to the actual materials were applied. As shown in Fig. 2 (c) and Table 1, the analytical results of the acoustic parameters were achieved by positioning the receiving point and sound source at the same point at a distance of 2–17 m, similar to the on-site measurements. On comparing the $L_{p,A,S,4m}$, $D_{2,S}$, and r_D values of the computer simulation model and the actual office space, a difference of less than 2% was recorded. Furthermore, the actual and simulation values for RT_{30} and D_{50} were identical, indicating that the computer simulation was modeled under practical conditions.

Based on the simulation model, we developed three models with different $L_{p,A,S,4m}$ values by selecting diverse values for the sound absorption coefficients of the ceiling and floor. The sound absorption coefficients of the interior flooring and ceiling materials were set to 0.5 and 0.05, and to 0.95 and 0.05, respectively, while maintaining the sound absorption coefficients for the walls and interior furniture in the computer simulation (see Appendix A). As shown in Fig. 2(c), $L_{p,A,S,4m}$ values of 45.9 (low), 51.1 (middle), and 55.9 (high) dB were achieved for the same sound source and receiver positions as in Fig. 2(b). The high absorptive ceiling and high absorptive floor were applied for the low condition, low absorptive ceiling and high absorptive floor for the

middle condition, and low absorptive ceiling and low absorptive floor for the high condition. The range of $L_{p,A,S,4m}$ values for a general OPO was considered with reference to the 43.4–57.2 dB range identified while examining the $L_{p,A,S,4m}$ distribution in 16 OPOs in the case study [60] presented in ISO 3382-3 Annex A (2012) and the 44.3–53.8 dB range identified while examining the $L_{p,A,S,4m}$ distribution in another set of 21 offices in a subsequent study [61]. This approach indicated a means of examining the subjective response to speech privacy by changing the sound source based on the various single number quantities presented in ISO 3382-3 (2012), including the $L_{p,A,S,4m}$ values in a previous study [51]. Consequently, a total of three room impulse responses were generated.

2.2.3. Background noise measurement

Although the room impulse response of the computer simulation model created previously included spatial sound field characteristics, it did not include any background noise information occurring at the office site. Because an actual office space would usually include background noise, the background noise should be obtained to generate an on-site working environment within the laboratory. Therefore, a sound level meter (SVAN 953, Svantek, Warsaw, Poland) was used to record background noise in an actual office space, similar to that chosen for the on-site measurement, for 1 h, while the space was occupied by workers. Fig. 3 shows the sound pressure level of the background noise while recording. The average L_{Aeq} value of the background noise was calculated to be 54.5 dB. This value is similar to the average background noise level of 53.6 dB obtained by Yadav et al. [62] in a case study in which the L_{Aeq} values of 43 OPOs while being occupied by workers were investigated. Furthermore, the background noise level was found to be moderate compared to those in previous studies [63–66]. A 1-min-long section within the recorded background noise sound source, in which speech-related noise (communication, speech, etc.) did not occur, was obtained; this section from the sound source was edited using Adobe Audition. Therefore, the selected section of the background noise sound source can be defined as the state in which the space is occupied by workers, but speech does not occur. Ultimately, the L_{Aeq} value of the 1-min background noise stimulus was found to be 47.0 dB, which was higher than the minimum L_{Aeq} level (45 dB) for performing collaborative, focused individual work at workstations, as presented in ISO 22955 [67]. Therefore, although it cannot be claimed that the background noise sound source selected for this study represented the general background noise level in an office, it did represent a poor office soundscape.

2.2.4. Audio stimuli generation

As shown in Fig. 4 (a), we used a 1 min male speech sample recorded in an anechoic chamber to generate high-quality irrelevant speech with audio stimuli and analyzed the frequency characteristics using the short-time Fourier transform. The speech sample had a relatively stronger frequency response below 500 Hz, which is consistent with the sound pressure level that lies in the range of 250–500 Hz, the highest band based on the sound power spectrum of normal speech as suggested by ISO 3382-3 [57]. The three-room impulse responses generated from the speech sample through computer simulation were convoluted. The three sound sources were then played on headphones and recorded using a head and torso simulator (Type 4100, Brüel & Kjær), while the sound pressure level was adjusted using Adobe Audition to correct the $L_{p,A,S,4m}$ values of each sound source, presented in Table 1. The 1 min background sound source was then added to the adjusted speech sound sources to generate three audio stimuli. In other words, the sound pressure level of the speech was changed into three different types, and the background noise level was fixed at 47 dB. The consequent signal-to-noise ratios of the stimuli were calculated to be −1.1 dB, 4.1 dB, and 8.9 dB. The duration of the evaluated stimuli was selected to be 1 min based on previous studies [68,69] in which office soundscape perception was evaluated using sound source stimuli lasting for more or

Table 1
Measurement and computer simulation results for the single-number quantities.

Variables	ISO 3382-3			ISO 3382-1	
	$L_{p,A,S,4m}$ [dB]	$D_{2,S}$ [dB]	r_D [m]	RT_{30} [s]	D_{50}
Measurement	49.4	4.6	11.1	0.6	0.89
Simulation	48.8	4.5	11.3	0.6	0.89
Low (L)	45.9	5.2	8.9	0.5	0.71
Middle (M)	51.1	3.9	13.2	0.7	0.67
High (H)	55.9	2.5	6.1	1.5	0.23

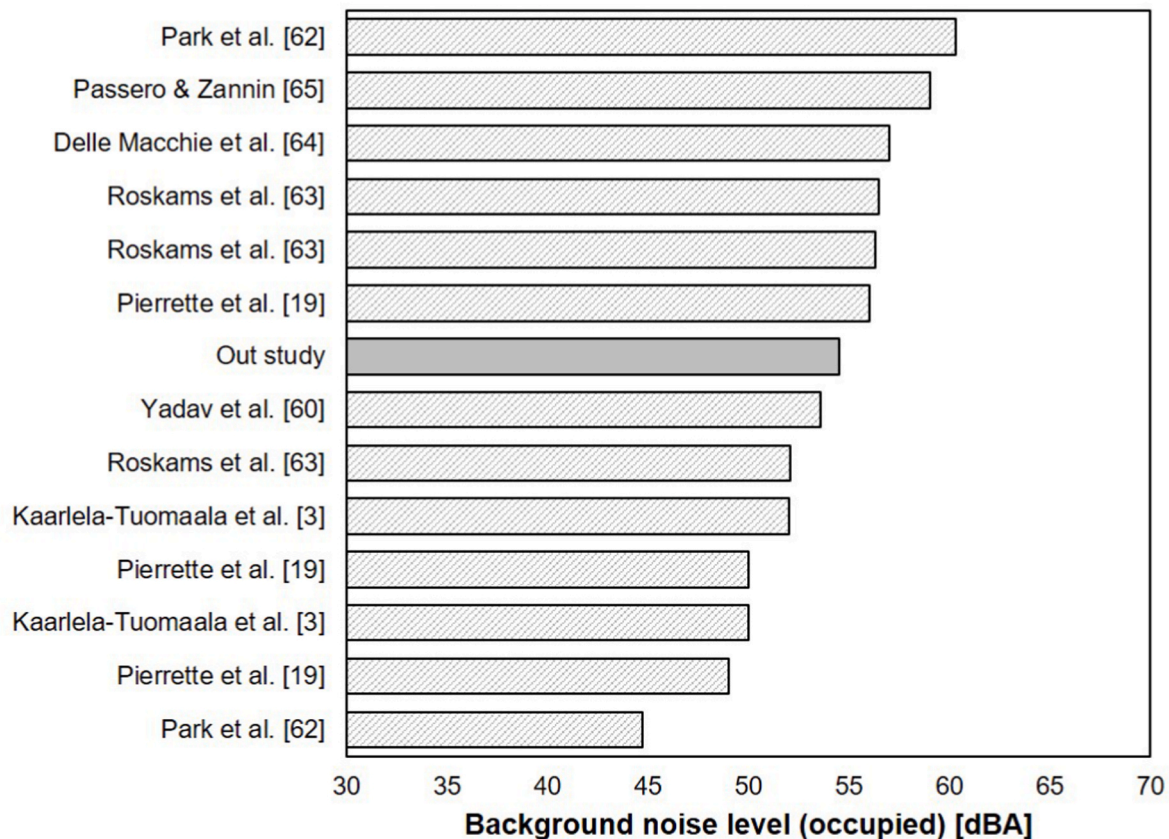


Fig. 3. Background noise levels of OPOs occupied by workers.

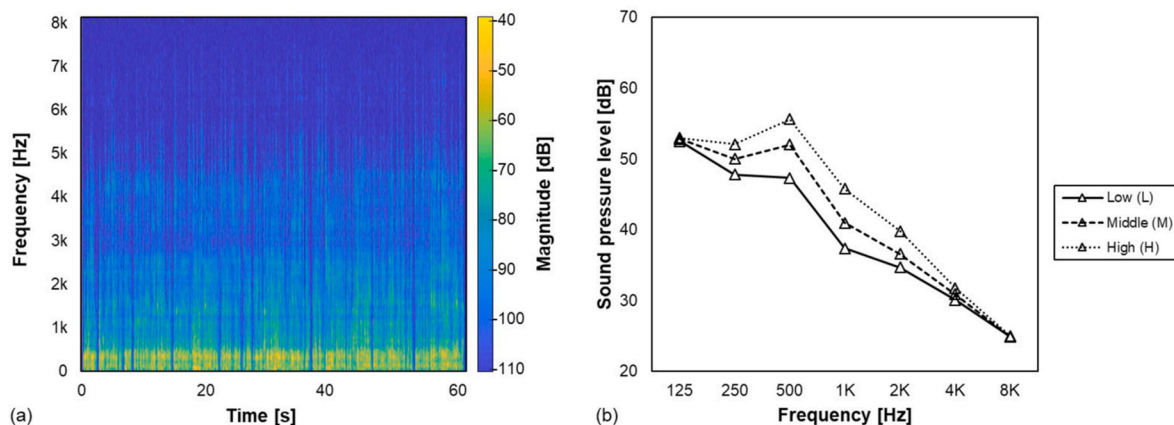


Fig. 4. (a) Spectral content of the male speech sample used in the audio stimuli generation. (b) Frequency characteristics of the three sound stimuli.

less than 1 min in a laboratory environment. Fig. 4(b) presents the results of the frequency responses of the three stimuli.

2.3. Visual stimuli generation using VR technology

We used VR technology, Unity 3D, a cross-platform game engine, to create four OPO spaces with visually varying internal conditions. Fig. 5 shows each OPO; their physical and visual characteristics are summarized in Table 2. We analyzed visual parameters such as the window-to-wall ratio (the ratio of the window area to the total wall area), W/L (the ratio of the window length to the ground width), the floor area (directly related to the size of the office that the worker visually faces), the capacity, and the floor area per person. The floor area per person refers to

the density of workers and is therefore related to the density of furniture such as workstations inside the office. W/L was selected as a visual parameter, considering that it represents the aspect ratio of space, making it relevant to the sense of space visually perceived by workers. Two types of window views were set up to provide a sense of the outside environment. Both windows faced the city with one providing a good view with a relatively pleasant urban space, and the other providing a bad view with a relatively underdeveloped and unpleasant urban space. The window area ratio was 10–45% inside the office, and the W/L was between 1.00 and 1.57. The floor area was measured to be between 57 and 585 m², and the area occupied by each occupant was between 10.6 and 24.2 m². Four office spaces were created to represent a general office ground plan and provide sufficient variations to determine the

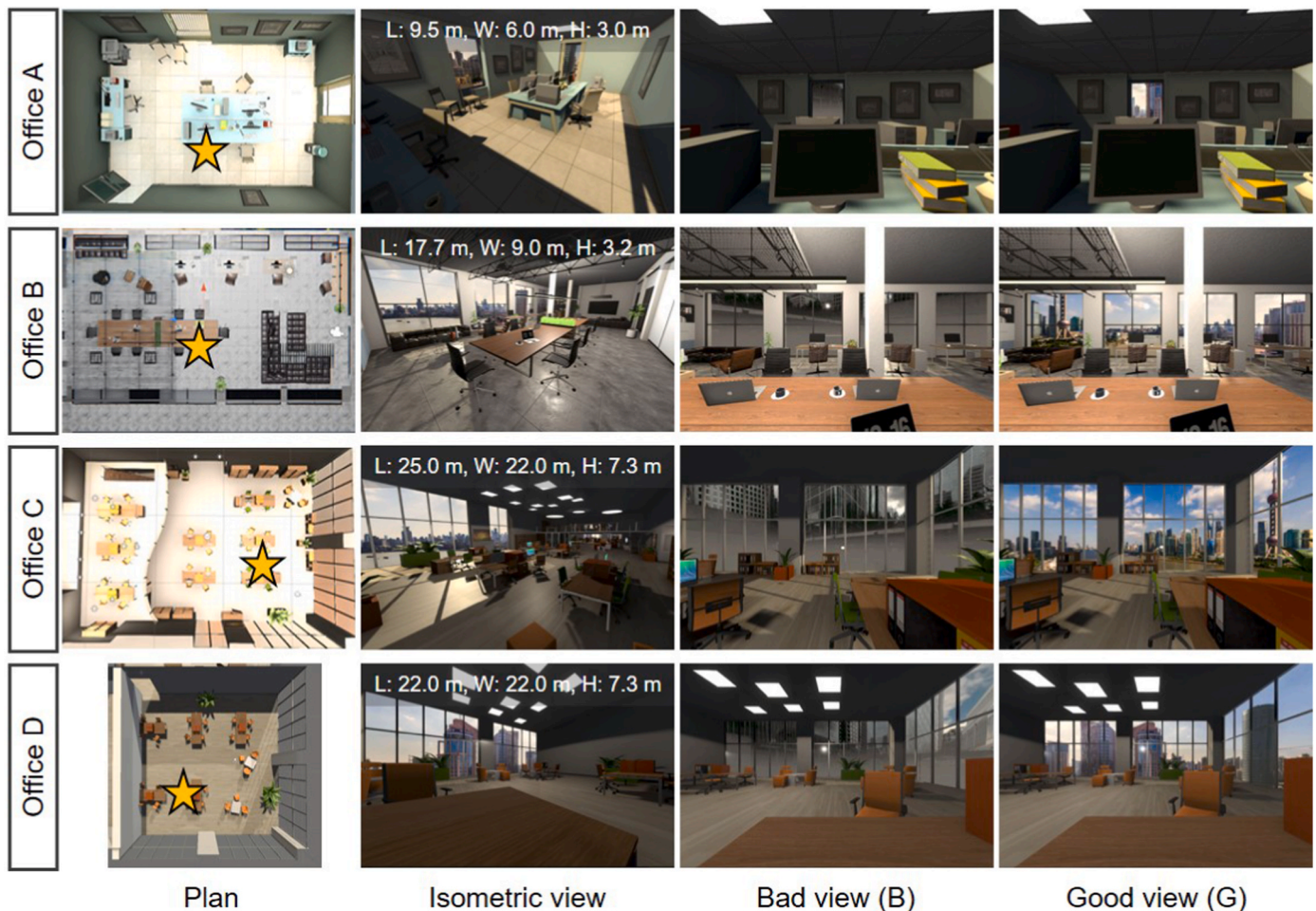


Fig. 5. View of the visual environments based on different OPO layouts and window views. The yellow stars indicate the experimental positions in the VR OPO environments. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2
Visual characteristics of different OPOs.

Variables	Office A	Office B	Office C	Office D
Window area ratio [%]	10	25	35	45
W/L	1.57	1.97	1.14	1.00
Floor area [m ²]	57	159	550	585
Capacity [N]	5	15	32	20
Floor area per person [m ²]	11.4	10.6	17.2	24.2

differences in the subjective responses based on visual changes. The positions were such that the subjects sat facing the nearby window, as shown in Fig. 5. Except for office A, which had a relatively smaller space, all the other office spaces had a distance of approximately 7 m between the window and assessment point. Furthermore, a function was provided for the subjects to move their heads and examine their surroundings freely while at their designated workstations. The viewing height was set as 1.2 m based on the average eye height of a person when seated; this coincided with the height of the receiver set in the computer simulation. To control visual influences other than architectural design factors, separate modeling was not performed for the people present in the working environment, considering the limitations that exist in generalizing the appearances or movements of people. A previous study [48] in which the audio-visual effects on the office soundscape were examined also did not include the visual aspects of people, despite speech occurring in this situation, and it was confirmed that the visual environment created was adequate for examining perceptive results.

Consequently, eight stimuli (four offices $\times$ two window views) were generated.

2.4. Experiment design

2.4.1. Questionnaire

The survey used for the subjective assessment of indoor soundscapes for OPOs comprised two sections. The first section asked the subjects to rate the extent to which each stimulus matched the semantic expression words for hearing (Loud, Variable, and Reverberant) and sight (Bright, Orderly, and Wide) on a 7-point Likert scale (−3: Not at all to 3: Extremely) to determine the perceived affective quality of the office environment. Each attribute was selected by referring to the acoustic and contextual factors. During selection, focus group interviews were conducted with five researchers specializing in indoor noise, to obtain seven attributes each for vision and hearing. Three attributes were ultimately selected to evaluate the intensity, change, and magnitude (sense of space) of the senses. Although the audio-visual attributes selected here were somewhat limited in representing all the indoor soundscapes of OPOs, this study is focused on evaluating various properties of the audio-visual environment rather than representing the general indoor soundscape since its purpose is to examine the crossed effect of audio-visual attributes.

The second section of the questionnaire determines the overall quality of the OPOs based on three aspects: preference, privacy, and work. Preference is classified into the overall acoustic and visual satisfactions, whereas privacy is classified in terms of speech and visual

privacy. The subjects were then asked to rate their preference for each assessment factor on a 7-point Likert scale (−3: Not at all; 3: Extremely). Lastly, the work attributes were classified into work satisfaction and task performance. Work satisfaction represents the satisfaction of a worker with the OPO environment, whereas task performance represents the actual productivity of workers based on the response accuracy for digit span tasks.

The auditory backward digit span task, a task in which subjects remember a sequence of given numbers and repeat them in the reverse order, was selected to evaluate short-term verbal memory in a short-term audio-visual working environment [70]. Starting with a sequence of two numbers, more numbers are added when the subject consecutively provides correct answers, and the sequence increases to eight numbers (2-2-3-4-5-6-7-8). The total section comprises 16 items. A sequence of numbers was provided to the subjects through a speaker, and the subjects provided the answers orally. The ratio of correct answers for all items was calculated based on the subject's responses and was quantified as task performance.

2.4.2. Procedure

We had 34 students from the same university (25 males and 9 females) with ages between 20 and 33 years (mean age = 24.85, standard deviation = 3.56) participate in this study. However, the number of subjects was set slightly higher at 34, considering 31 samples were used to set a testing power of 0.8 and effect size of 0.35. The sample size was calculated using G-Power software (version 3.1.9.4). The subjects were enlisted based on their experience of working in an office space. The subjects were asked to sign a consent form before the experiment with respect to the ethical guidelines. They were given instructions on how to use the VR device, and exercise sessions were conducted for the digit span task to ensure the subjects understood the task well. The assessment was conducted in a quiet laboratory with a significantly low background noise of approximately 30 dBA. Furthermore, the inner wall of the laboratory was finished with sound absorption panels composed of light-brown porous wood.

The experimental procedure was as follows. Firstly, each subject was asked to wear open-type headphones (Sennheiser HD 650) and a head-mounted display (VIVE Pro EYE). Then, 24 stimuli (three audio stimuli × visual stimuli) were presented randomly. Subsequently, a 1-min-long VR audio-visual office stimulus was presented to each subject. During this process, each subject was allowed to move their head and eye freely. After the 1-min-long experience, a survey was conducted in the form of a GUI presented within the VR, and each subject was asked to respond with a controller. The same VR audio-visual office stimulus was repeated to ensure that each subject fully experienced and understood the space. Therefore, the total time spent providing survey responses was different for each subject, owing to their various adaptation times to the audio-visual stimulus. Once the survey had been completed, a digit span task was performed, where the same stimulus was repeatedly experienced by each subject. When all responses to one stimulus had been received, the next stimulus was presented after approximately 30 s of rest time. By repeating this process, 816 responses (24 stimuli × 34 subjects = 816) were collected for each item.

2.5. Data analysis

2.5.1. Normalization

Owing to individual differences (discrepancy in interpretation, sensitivity, and intelligence level), the assessment standards for each subject may differ even when using the same scale and task. Therefore, the range of responses should be normalized before determining the mean value [71]. The normalized answer of person p for stimulus i and question j is obtained as follows [72]:

$$X_{norm,i,j,p} = X_{i,j,p} \sqrt{\frac{\sum_{i,j} X_{i,j}^2}{\sum_{i,j} X_{i,j,p}^2}}, \quad (1)$$

where $X_{i,j,p}$ is the initial answer and $X_{norm,i,j,p}$ is the normalized answer of person p for stimulus i and question j , $\sum_{i,j} X_{i,j,p}^2$ is the sum of the squares of

all answers from person p , $\overline{\sum_{i,j} X_{i,j}^2}$ is the average of the sum of the squares

for all subjects, where $\overline{\sum_{i,j} X_{i,j}^2} = \sum_{p=1}^n \left(\frac{1}{p} \right) \sum_{i,j} X_{i,j,p}^2$.

2.5.2. Statistical analysis

The following analyses were conducted using the SPSS Statistics. Because the Kolmogorov–Smirnov and Shapiro–Wilk tests were conducted on the survey results, the significance level was lower than 0.05, which did not satisfy the normality. Therefore, nonparametric statistics were applied. Furthermore, the Friedman and Wilcoxon signed-rank tests were performed to analyze the median difference owing to the audio-visual conditions of each stimulus. Additionally, Spearman correlation analysis was conducted to determine the correlation between the physical soundscape characteristics of an OPO and the normalized subjective data (perceived affective quality and overall quality), based on which the soundscape design methodology was presented for a pleasant indoor soundscape design in OPOs.

3. Results

3.1. Crossed effect of the audio-visual physical parameter on the perceived affective quality

Fig. 6 shows the mean responses for the perceived affective quality of the audio-visual environment in OPOs. 24 stimuli were ranked and arranged in descending order based on the overall preference values. The Spearman correlation coefficients r_s between the perceived affective quality and overall preference were analyzed. The sound attribute results indicate that the overall preference has negative correlations with the perception of the Loud ($r_s = -0.54$), Variable ($r_s = -0.66$), and Reverberant ($r_s = -0.60$) attributes, as shown in Fig. 6(a). Conversely, the visual attribute results indicate that the overall preference has positive correlations with the perception of the Bright ($r_s = 0.73$), Orderly ($r_s = 0.85$), and Wide ($r_s = 0.81$) attributes, as shown in Fig. 6(b).

Furthermore, we conducted the Friedman and Wilcoxon signed-rank tests to determine the significance of the median differences in the responses by the subjects on the perceived affective quality based on three acoustic conditions ($L_{p,A,S,4m}$), four indoor visual environments (office layouts), and two outdoor visual environments (window views). For Loud, $L_{p,A,S,4m}$ presented medians of 2.81 for high (55.9 dB), 1.91 for middle (51.1 dB), and 1.27 for low (43.9 dB). Furthermore, the Friedman test results indicated that the differences in loudness across the $L_{p,A,S,4m}$ values were statistically significant ($p < 0.01$), which could result in a high response rate for Loud. Similarly, we analyzed the median changes based on different $L_{p,A,S,4m}$ values for the Variable and Reverberant responses and found that smaller $L_{p,A,S,4m}$ values decreased the Variable and Reverberant median in the order of 2.62 (H) – 1.87 (M) – 1.51 (L) and 2.89 (H) – 1.86 (M) – 1.25 (L), respectively. Among all visual attributes, Bright and Wide did not show a statistically significant median change with respect to changes in the $L_{p,A,S,4m}$ values. However, the median for Orderly was 1.90 (H) – 1.96 (M) – 2.15 (L), indicating that the response median at an $L_{p,A,S,4m}$ of 43.9 dB was the highest and that Orderly was likely to be rated higher for low $L_{p,A,S,4m}$ values.

Median changes in the perceived affective quality based on the four types of indoor visual environments were recorded as follows. The median for Bright was 1.36 (A), 2.91 (B), 2.78 (C), and 2.95 (D), and the difference across the visual conditions was statistically significant ($p <$

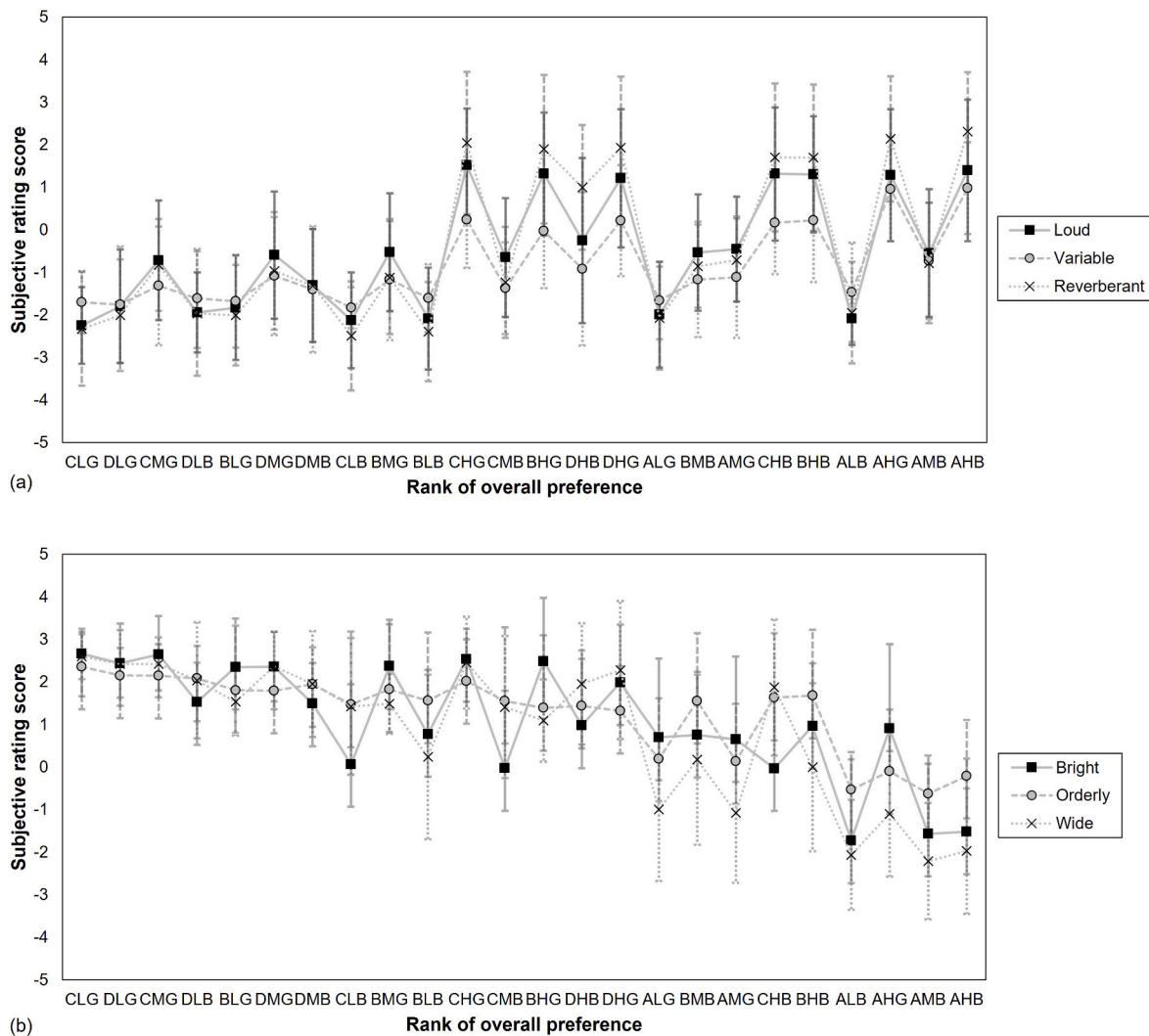


Fig. 6. Mean ratings of perceived affective quality for different OPO environments based on overall preference, in descending order. (a) Acoustic attributes and (b) visual attributes. The first, second, and third letters represent the office type, $L_{p,A,S,4m}$ (L: low, M: middle, H: high), and window view (B: bad view, G: good view), respectively. [Appendix B](#) provides the detailed values.

0.01). The median for Orderly and Wide were 1.56 (A) – 2.66 (B) – 2.77 (D) – 3.01 (C) and 1.12 (A) – 2.40 (B) – 3.24 (C and D), respectively, indicating that the difference in the median of Orderly and Wide across the visual conditions was significant ($p < 0.01$). Furthermore, a significant change was observed in the sound attributes. The response medians for Loud, Variable, and Reverberant decreased as D-C-B-A when the office layout changed, with a maximum difference of less than 1, which is smaller than the scale on the Likert scale; therefore, the effect was minimal. Lastly, the response changes in the perceived affective quality based on the two types of outdoor visual environments were analyzed using the same approach as that for the indoor visual environments. We conducted the Wilcoxon signed-rank to determine whether the response changes were statistically significant for the good and bad views. Results showed that Bright, Orderly, and Wide presented higher response rates for the good view compared to the bad view. Furthermore, similar to the visual attributes, the responses for Loud and Reverberant were higher in the good view.

To summarize the relationships between the crossed effects of the audio-visual physical parameters and perceived affective quality, which was the first issue addressed in this study, the physical sound indices of the office were only related to the perceived affective quality, with almost no crossed effect concerning the visual environment. Meanwhile, the physical visual indices were significantly related to the perceived

affective quality concerning the soundscape, confirming the crossed effect. Therefore, it can be concluded that it is necessary to consider the acoustic and visual factors when evaluating the indoor soundscape of a working environment.

Based on the above results, we conducted the Spearman correlation analysis to analyze the correlation between the physical parameters and the perceived affective quality. [Table 3](#) summarizes the correlation coefficients. The window view was designated as a dummy variable, and the good and bad views were preprocessed as 1 and 0, respectively. While $L_{p,A,S,4m}$ and RT_{30} showed a strong positive correlation with the Loud ($r_s = 0.68$), Variable ($r_s = 0.49$), and Reverberant ($r_s = 0.76$) attributes, $D_{2,S}$ and D_{50} showed a strong negative correlation [Loud ($r_s = -0.68$), Variable ($r_s = -0.49$), and Reverberant ($r_s = -0.76$)]. Furthermore, r_D showed a moderate negative correlation with the Loud ($r_s = -0.33$), Variable ($r_s = -0.30$), and Reverberant ($r_s = -0.47$) attributes. However, no statistically significant correlation was observed between the visual and acoustic parameters. Subsequently, among all visual parameters, the outdoor window view of the OPOs had a strong positive correlation with the Bright ($r_s = 0.51$) attribute and a weak positive correlation with the Wide ($r_s = 0.25$) attribute. Additionally, the window ratio and floor area showed a clear positive correlation with the Bright ($r_s = 0.35$), Orderly ($r_s = 0.38$), and Wide ($r_s = 0.66$) attributes. Moreover, while the floor area per person showed a positive correlation

Table 3Spearman correlation coefficient (r_s) between the physical parameters and the perceived affective quality (* $p < 0.05$, ** $p < 0.01$).

Domain	Auditory			Visual		
	Loud	Variable	Reverberant	Bright	Orderly	Wide
Acoustic parameter						
$L_{p,A,S,4m}$ [dB]	0.68**	0.49**	0.76**	-0.01	-0.07	-0.01
$D_{2,S}$ [dB]	-0.68**	-0.49**	-0.76**	0.01	0.07	0.01
r_D [m]	-0.33**	-0.30**	-0.47**	0.01	0.03	-0.02
RT_{30} [s]	0.68**	0.49**	0.76**	-0.01	-0.07	-0.01
D_{50}	-0.68**	-0.49**	-0.76**	0.01	0.07	0.01
Visual parameter						
View	0.05	0.01	0.04	0.51**	0.04	0.25**
Window area ratio [%]	-0.08*	-0.14**	-0.07*	0.35**	0.38**	0.66**
W/L [%]	0.07*	0.09*	0.04	-0.11**	-0.17**	-0.45**
Floor area [%]	-0.08*	-0.14**	-0.07*	0.35**	0.38**	0.66**
Floor area per person [m^2]	-0.07*	-0.09*	-0.04	0.11**	0.17**	0.45**

with the Wide ($r_s = 0.45$) attribute, the W/L ratio showed a negative correlation [Wide ($r_s = -0.45$)]. However, no statistically significant correlation was observed between the acoustic and visual parameters concerning the audio-visual crossed effect.

3.2. Crossed effect of the audio-visual physical parameter on the overall quality

Fig. 7 shows the results of the overall quality assessment of OPOs under different audio-visual environmental conditions. As mentioned previously, the overall preference for each stimulus was arranged in descending order. All assessment indicators except task performance ($r_s = -0.13$), such as acoustic satisfaction ($r_s = 0.79$), visual satisfaction ($r_s = 0.88$), speech privacy ($r_s = 0.71$), visual privacy ($r_s = 0.56$), and work satisfaction ($r_s = 0.98$), showed a strong positive correlation with the overall preference.

Similar to the analysis method for the perceived affective quality, we conducted the Friedman and Wilcoxon signed-rank tests to determine any significance in the median difference of responses concerning the overall quality based on the different acoustic and indoor/outdoor visual environment conditions. The median changes in the overall quality with respect to the changes in the $L_{p,A,S,4m}$ values indicated that the acoustic satisfaction had changed significantly for different $L_{p,A,S,4m}$, and speech privacy values in the order of 1.26 (H) – 2.19 (M) – 2.55 (L) and

1.31 (H) – 2.13 (M) – 2.56 (L), respectively. Furthermore, as the $L_{p,A,S,4m}$ value for work satisfaction decreased, the response median changed in the order of 1.48 (H) – 2.13 (M) – 2.39 (L), and the median for task performance also decreased in the order of 1.75 (H) – 1.98 (M) – 2.27 (L). Conversely, the effect of $L_{p,A,S,4m}$ on visual satisfaction ($p = 0.15$) and visual privacy ($p = 0.21$) were statistically insignificant. Therefore, to improve the overall sound quality, it is desirable to provide an office soundscape with a lower $L_{p,A,S,4m}$ value.

Furthermore, the effect of indoor visual environments showed that the visual satisfaction increased in the order of 1.36 (A) – 2.53 (B) – 3.02 (C) – 3.09 (D) as the office layout space increased. Visual privacy showed a high response median in the order of 2.07 (B) – 2.41 (A) – 2.56 (D) – 2.97 (C), indicating a slightly different tendency from visual satisfaction. Furthermore, the indoor visual attributes exhibited a significant change in the perception of the audio-related overall quality. The acoustic satisfaction presented a similar tendency to visual satisfaction in the order of 1.93 (A) – 2.44 (B) – 2.71 (C) – 2.92 (D) across different office layouts, whereas speech privacy presented the response median distribution in the order of 2.22 (A) – 2.49 (B) – 2.62 (D) – 2.68 (C). Therefore, with adequate indoor visual factors, a significant difference can be seen in both visual preference and auditory assessment of space. Conversely, work satisfaction and task performance showed a response median change in the order of 1.45 (A) – 2.61 (B) – 2.91 (C) – 3.03 (D) and 1.89 (D) – 2.49 (C) – 2.68 (B) – 2.93 (A), respectively, for

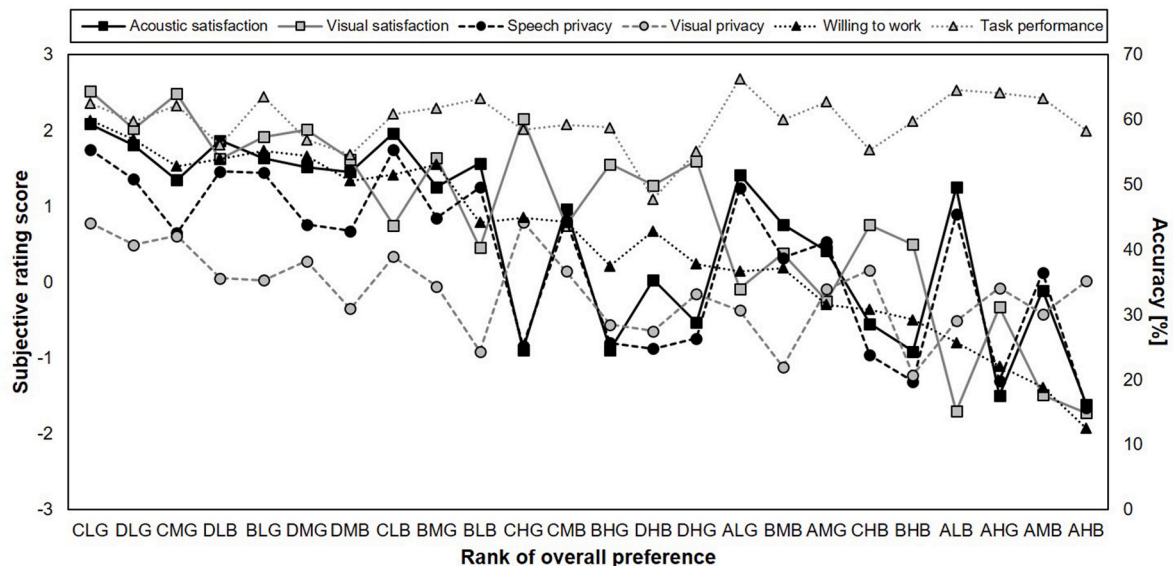


Fig. 7. Mean rating score of the overall quality for different OPO environments based on the overall preference, in descending order. The first, second, and third letters indicate the office type, $L_{p,A,S,4m}$ (L: low, M: middle, H: high), and window view (B: bad view, G: good view), respectively. Appendix B provides the detailed values.

different office layouts.

While work satisfaction represented the psychological aspect, task performance represented actual work productivity. However, no significant correlation was observed for either indicator ($r_s = 0.15$) in the Spearman correlation analysis. Therefore, it is necessary to provide an OPO design that balances these two indicators based on the use of space (whether the focus is on the workers' concentration or the visitors' and users' awareness of space). Lastly, the response changes in the overall quality based on the changes in the outdoor visual environments indicated that visual satisfaction, visual privacy, and speech privacy showed higher rates in the good view than in the bad view. Likewise, the work satisfaction and task performance ratings were higher with the good view than with the bad view.

Spearman correlation analysis was conducted to analyze the correlation between the physical parameters and the overall quality, and the results are summarized in Table 4. Acoustic satisfaction showed a strong negative correlation ($r_s = -0.57$) with $L_{p,A,S,4m}$ and RT_{30} , a strong positive correlation ($r_s = 0.57$) with $D_{2,S}$ and D_{50} , and a positive correlation with r_D ($r_s = 0.36$). Similarly, visual satisfaction showed a clear positive correlation ($r_s = 0.49$) with the window area ratio and floor area, a negative correlation with the W/L ratio ($r_s = -0.29$), and a weak positive correlation with the view ($r_s = 0.32$) and floor area per person ($r_s = 0.29$).

Speech privacy showed a similar tendency to acoustic satisfaction, showing a strong negative correlation ($r_s = -0.54$) for $L_{p,A,S,4m}$ and RT_{30} , and a clear positive correlation ($r_s = 0.54$) with $D_{2,S}$, D_{50} , and r_D ($r_s = 0.33$). Conversely, visual privacy showed a negative correlation with W/L ($r_s = -0.23$), a positive correlation with the floor area per person ($r_s = 0.16$) and view ($r_s = 0.15$), indicating a similar tendency with visual satisfaction, and an extremely weak correlation ($r_s = 0.10$) with the window area ratio and floor area. Therefore, the overall correlation remains unclear.

Lastly, in terms of work attributes, work satisfaction showed a clear negative correlation ($r_s = -0.34$) with $L_{p,A,S,4m}$ and RT_{30} and a clear positive correlation ($r_s = 0.34$) with $D_{2,S}$ and D_{50} . Among the visual parameters, a positive correlation was observed between the window area ratio and floor area ($r_s = 0.42$), floor area per person ($r_s = 0.23$), and view ($r_s = 0.21$), whereas a weak negative correlation was observed with W/L ($r_s = -0.23$). Task performance showed a negative correlation ($r_s = -0.21$) with $L_{p,A,S,4m}$ and RT_{30} among the acoustic parameters and a positive correlation ($r_s = 0.21$) with $D_{2,S}$ and D_{50} , showing a similar tendency as work satisfaction. Among the visual parameters, a negative correlation was observed between the window area ratio and floor area ($r_s = -0.29$) and the floor area per person ($r_s = -0.25$), and a positive correlation between the W/L ratio ($r_s = 0.25$) and view ($r_s = 0.12$). In summary, all visual parameters except window view showed an entirely opposite relationship with work satisfaction and task performance.

Therefore, to summarize the relationships between the crossed effects of the audio-visual physical parameters and overall quality, which

was the second topic investigated in this study, the physical sound parameters were significantly related to the overall quality of the soundscape, including acoustic satisfaction and speech privacy, similar to the perceived affective quality. Meanwhile, the physical visual parameters were significantly related to visual satisfaction, visual privacy, and the overall quality of the soundscape, indicating a crossed effect.

3.3. Crossed effect of audio-visual attributes on overall quality

We analyzed the Spearman correlation coefficients between the two assessment factors to evaluate the audio-visual crossed effect based on the subjective responses of the subjects, to the perceived affective quality and the overall quality. The results are presented in Table 5. Acoustic satisfaction showed a strong negative correlation with the Loud ($r_s = -0.61$), Variable ($r_s = -0.61$), and Reverberant ($r_s = -0.64$) attributes, and a weak positive correlation with the Bright ($r_s = 0.20$), Orderly ($r_s = 0.22$), and Wide ($r_s = 0.24$) attributes. Conversely, visual satisfaction showed a strong positive correlation with the Bright ($r_s = 0.64$), Orderly ($r_s = 0.58$), and Wide ($r_s = 0.65$) attributes, and an extremely weak negative correlation with the Loud ($r_s = -0.10$), Variable ($r_s = -0.21$), and Reverberant ($r_s = -0.09$) attributes.

Speech privacy showed a negative correlation with the Loud ($r_s = -0.51$), Variable ($r_s = -0.50$), and Reverberant ($r_s = -0.57$) attributes similar to acoustic satisfaction, and a weak positive correlation with the Bright ($r_s = 0.10$), Orderly ($r_s = 0.21$), and Wide ($r_s = 0.15$) attributes. Visual privacy showed a slightly different tendency from visual satisfaction, showing a weak positive correlation with the Bright ($r_s = 0.13$), Orderly ($r_s = 0.25$), and Wide ($r_s = 0.19$) attributes, and a partial, but extremely weak negative correlation with the Variable ($r_s = -0.09$) attribute.

Both work satisfaction and task performance showed negative correlations with the Loud ($r_s = -0.35$ and $r_s = -0.18$ respectively), Variable ($r_s = -0.43$ and $r_s = -0.12$ respectively), and Reverberant ($r_s = -0.40$ and $r_s = -0.19$ respectively) attributes. However, correlations with the visual attributes were different. Work satisfaction showed a clear positive correlation with the Bright ($r_s = 0.47$), Orderly ($r_s = 0.47$), and Wide ($r_s = 0.53$) attributes, whereas task performance showed a negative correlation with the Wide ($r_s = -0.15$) attribute.

Therefore, the relationship between the crossed effect of audio-visual perceived affective quality and overall quality, which was the last research topic examined in this study, can be summarized as follows. Unlike the physical sound indices examined earlier, the acoustic attributes are significantly related to the overall quality of the visual environment, although the relationship is not significant, indicating the presence of a crossed effect. Furthermore, although the acoustic attributes show negative yet similar relationships with work satisfaction and task performance, the visual attributes exhibit opposite relationships with work performance and task performance. Therefore, when designing a workspace, the visual environment must be carefully

Table 4

Spearman correlation coefficient (r_s) between the physical parameters and the overall quality (* $p < 0.05$, ** $p < 0.01$).

Domain	Satisfaction		Privacy		Work	
	Acoustic satisfaction	Visual satisfaction	Speech privacy	Visual privacy	Work satisfaction	Task performance
Attributes						
Acoustic parameter						
$L_{p,A,S,4m}$ [dB]	-0.57**	-0.05	-0.54**	-0.05	-0.34**	-0.21**
$D_{2,S}$ [dB]	0.57**	0.05	0.54**	0.05	0.34**	0.21**
r_D [m]	0.36**	0.04	0.33**	0.03	0.22**	0.11**
RT_{30} [s]	-0.57**	-0.05	-0.54**	-0.05	-0.34**	-0.21**
D_{50}	0.57**	0.05	0.54**	0.05	0.34**	0.21**
Visual parameter						
View	0.03	0.32**	0.05	0.15**	0.21**	0.12**
Window area ratio [%]	0.21**	0.49**	0.10**	0.10**	0.42**	-0.29**
W/L [%]	-0.14**	-0.29**	-0.06	-0.16**	-0.23**	0.25**
Floor area [%]	0.21**	0.49**	0.10**	0.10**	0.42**	-0.29**
Floor area per person [m ²]	0.14**	0.29**	0.06	0.16**	0.23**	-0.25**

Table 5Spearman correlation coefficient (r_s) between the perceived affective quality and the overall quality ($*p < 0.05$, $**p < 0.01$).

Domain	Satisfaction		Privacy		Work	
Attributes	Acoustic satisfaction	Visual satisfaction	Speech privacy	Visual privacy	Work satisfaction	Task performance
Acoustic						
Loud	−0.61**	−0.10**	−0.51**	−0.03	−0.35**	−0.18**
Variable	−0.61**	−0.21**	−0.50**	−0.09*	−0.43**	−0.12**
Reverberant	−0.64**	−0.09**	−0.57**	−0.03	−0.40**	−0.19**
Visual						
Bright	0.20**	0.64**	0.10**	0.13**	0.47**	0.01
Orderly	0.22**	0.58**	0.21**	0.25**	0.47**	−0.04
Wide	0.24**	0.65**	0.15**	0.19**	0.53**	−0.15**

considered to ensure a balance between work performance and task performance.

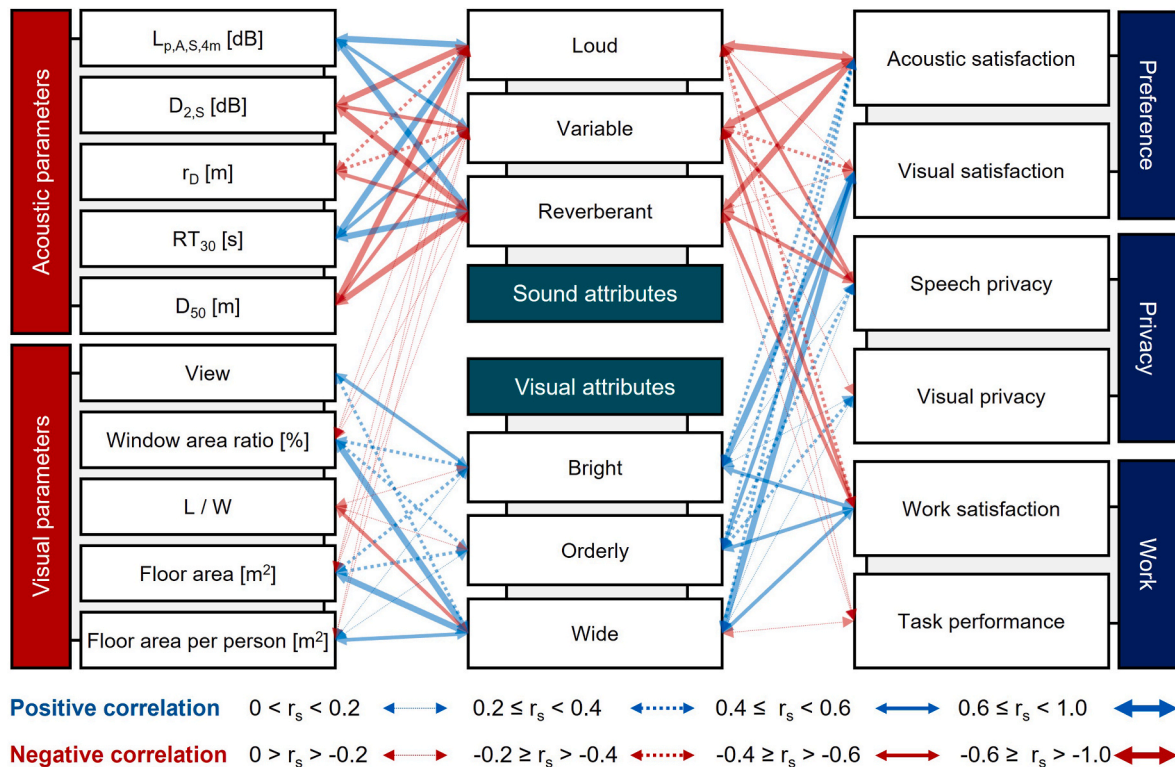
4. Discussion

4.1. Crossed effect of the audio-visual environment for a pleasant OPO indoor soundscape

Based on the analysis of the audio-visual crossed effect in OPOs, this study presented an indoor soundscape perception model (see Fig. 8) and proposed new methodologies for designing indoor soundscapes in OPOs for improving overall work satisfaction, privacy, and performance. We found that creating a visually bright, spacious, and well-organized space with a dry, acoustically calm environment having minimal reverberation was highly effective in improving the overall quality of the OPO. One can refer to the criteria for ensuring the minimum acceptable sound performance based on the six types of OPOs proposed in ISO 22955 [67]. These results can be obtained using various methods, such as by applying sound-absorbing finish to office interiors (ceiling, floor, and walls), installing sound absorption objects or partitions, and performing layout modification [7]. In particular, a significant change was observed in the overall quality owing to the $L_{p,A,S,4m}$ or $D_{2,S}$ values. Therefore, a

$L_{p,A,S,4m}$ value below 48 dB and $D_{2,S}$ higher than 7 dB is desirable for designing a workspace [57,60]. Furthermore, designing a bright, wide, and organized indoor space can be vital for visual satisfaction and visual privacy. This can be achieved by providing a good view from the office windows. However, if the office location cannot be changed, increasing the openness in the space by enlarging the windows or appropriately maintaining a low number of occupants can be beneficial to ensure that each worker has sufficient space. Previous studies [53,73] have reported that providing a window ratio of approximately 60% has been most advantageous in improving task performance. Additionally, improving the visual design can positively affect the perception of noise and variations in the acoustic aspect. Therefore, visual methods must be actively used. However, compared to acoustic parameters, if the physical attributes of OPOs are altered significantly, no significant change occurs in the overall quality of visual parameters. In other words, although visual privacy is essential for determining satisfaction in OPOs [6], no clear correlation was observed with the physical attributes of an office in this study. This can be due to the lack of consideration of other physical parameters (window proximity or partition height) attributed to visual privacy. Therefore, further research regarding these attributes is necessary [74].

The two major findings of this study are as follows. First, although

**Fig. 8.** Indoor soundscape perception model for OPO based on Spearman correlation coefficient.

the overall preference for space shows a positive correlation with work satisfaction, it does not exhibit a significant correlation with task performance. This finding is similar to the result obtained by Haka [18], who reported that occupant comfort is more significantly affected by the surrounding environment than performance. Essentially, the work satisfaction of subjects responded more sensitively and positively with the improvement in audio-visual conditions. Second, visual parameters in the indoor environment showed opposite correlations with work satisfaction and task performance. For the outdoor visual parameters, a pleasant view outside the office can improve work satisfaction and task performance, considering that the office is located in a well-planned urban landscape. However, if work satisfaction is given priority in terms of the indoor visual parameters, increasing the ratio of windows on the walls to provide a more occupied space for each worker can increase the workers' satisfaction. Conversely, in terms of task performance, reducing the ratio of windows on the walls to provide a relatively small space to decrease the area occupied by each worker can increase work productivity. This finding is partially consistent with the results obtained by Torresin et al. [75], who claimed that a private and controlled environment should be created in a space in which work-related activity occurs, unlike in a space in which relaxations are required. Despite this contrary tendency, work satisfaction positively contributes to task performance [76]. Therefore, designers must balance satisfaction and productivity from both employer and employee perspectives and consider both the workers' concentration as well as visitors' or users' perceptions of the space.

4.2. Study limitations

Despite the positive results obtained, this study has several limitations. First, only subjects in their 20s participated in this study, normality was not achieved, and only visual and audio stimuli in a limited office space were used, which limits the possibility of obtaining a clear cognitive model. Furthermore, it was not possible to achieve a gender balance in the recruited subjects. Although task performance was quantified with the digit span test, various cognitive skills used in actual tasks should be considered [77]. In addition, although short-term stimuli were used in this study to examine the responses of subjects to the changes in different environmental conditions, the various long-term effects (fatigue, stress, etc.) that could occur owing to the office environment were not considered. Therefore, a generalizable model that can be utilized directly when designing an office space must be derived from longitudinal studies in the future [78]. Since the audio-visual attributes evaluated in this study were selected through a small focus group interview, they do not represent the indoor soundscape of a general OPO. As a measure to evaluate the indoor soundscape is being proposed recently in a study [35], it is necessary to study the evaluation scale representing the indoor soundscape of an OPO through a follow-up study. In addition, this study was limited to visual and auditory senses, and the crossed effects with other indoor environmental quality

factors (air, thermal, etc.) should also be analyzed [8]. Moreover, only male sound sources were used in this study, i.e., female sound sources with different fundamental frequencies were not considered [79]. Lastly, further research is required to determine the effects of individual worker characteristics such as sensitivity, work satisfaction, career, gender, education, and economic level [80].

5. Conclusions

In this study, the relationships between various audio-visual characteristics and subjective assessments were analyzed to create a means of achieving a pleasant OPO soundscape that can be used when designing indoor soundscapes. The findings of this study, presented below, can be used as basic information for designing indoor soundscapes in OPOs.

1. Although the crossed effect of acoustic factors on the overall quality of the visual environment in a workspace was not substantial, it was significant for the visual factors. Therefore, the visual crossed effect must be considered when evaluating workplace soundscapes.
2. Increasing visual parameters such as the window area ratio and floor area can improve the overall quality of the visual environment and the audio satisfaction and speech privacy through the audio-visual crossed effect.
3. Although providing minimal acoustical reverberation and a calm environment can ensure work satisfaction and task performance, the effects of visual factors on work satisfaction and task performance were found to be contradictory. Therefore, when designing a workspace, a visual environment design must be developed considering a balance between work satisfaction and task performance based on the needs of the client.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Absorption coefficients of different materials for modeling using Odeon

No.	Element	Materials	Frequency								
			63	125	250	500	1000	2000	4000	8000	a
1	Desk, Drawer, Partition	1-cm-thick plywood paneling (Harris, 1991)	0.28	0.28	0.22	0.17	0.09	0.1	0.11	0.11	0.10
2	Wall	Brick, unglazed, painted (Harris, 1991)	0.01	0.01	0.01	0.02	0.02	0.02	0.03	0.03	0.05
3-1	Floor	Linoleum or vinyl stuck to concrete (Petersen, 1983)	0.02	0.02	0.02	0.03	0.04	0.04	0.05	0.05	0.05
3-2	Floor	Carpet, heavy, on hair felt or foam rubber (Harris, 1991)	0.08	0.08	0.24	0.57	0.69	0.71	0.73	0.73	0.50
4-1	Ceiling	Plaster, gypsum, or lime, smooth finish on lath (Harris, 1991)	0.14	0.14	0.1	0.06	0.04	0.04	0.03	0.03	0.05
4-2	Ceiling	100 mm thick wood-wool against the ceiling (Ref. Dalenback, CATT)	0.58	0.58	0.69	0.96	0.97	0.99	0.91	0.91	0.95

Appendix B. Subjective responses according to different audio-visual OPO environments in descending order of overall preference

Case	ID	Office	View	L _{p,A,S,4m}	OP	Perceived affective quality						Overall quality					
						Bright	Orderly	Wide	Loud	Variable	Reverberant	AS	VS	SP	VP	WS	TP
1	CLG	C	Good	45.9	2.32	2.66	2.36	2.59	-2.25	-1.70	-2.33	2.09	2.52	1.75	0.78	2.14	62.57
2	DLG	D	Good	45.9	2.14	2.44	2.15	2.42	-1.80	-1.76	-2.01	1.81	2.02	1.36	0.49	1.88	59.76
3	CMG	C	Good	51.1	1.93	2.64	2.14	2.42	-0.72	-1.32	-0.83	1.35	2.49	0.65	0.61	1.53	62.20
4	DLB	D	Bad	45.9	1.90	1.52	2.08	2.04	-1.94	-1.61	-1.96	1.87	1.62	1.46	0.05	1.62	56.07
5	BLG	B	Good	45.9	1.80	2.35	1.80	1.54	-1.83	-1.68	-2.01	1.64	1.92	1.44	0.02	1.73	63.54
6	DMG	D	Good	51.1	1.75	2.36	1.79	2.36	-0.59	-1.08	-0.97	1.52	2.01	0.76	0.28	1.66	56.90
7	DMB	D	Bad	51.1	1.63	1.49	1.95	1.95	-1.31	-1.40	-1.32	1.46	1.62	0.67	-0.35	1.34	54.57
8	CLB	C	Bad	45.9	1.56	0.06	1.47	1.42	-2.13	-1.83	-2.49	1.96	0.75	1.74	0.34	1.42	60.88
9	BMG	B	Good	51.1	1.49	2.37	1.83	1.49	-0.53	-1.17	-1.12	1.25	1.64	0.85	-0.05	1.55	61.80
10	BLB	B	Bad	45.9	1.12	0.77	1.56	0.24	-2.09	-1.60	-2.39	1.57	0.46	1.25	-0.91	0.79	63.24
11	CHG	C	Good	55.9	0.97	2.53	2.02	2.45	1.52	0.24	2.04	-0.89	2.15	-0.84	0.79	0.85	58.46
12	CMB	C	Bad	51.1	0.94	-0.03	1.55	1.41	-0.65	-1.37	-1.23	0.96	0.75	0.81	0.14	0.79	59.22
13	BHG	B	Good	55.9	0.73	2.48	1.39	1.10	1.31	-0.02	1.90	-0.89	1.55	-0.80	-0.56	0.22	58.82
14	DHB	D	Bad	55.9	0.71	0.98	1.44	1.95	-0.25	-0.92	0.99	0.03	1.28	-0.88	-0.64	0.68	47.77
15	DHG	D	Good	55.9	0.62	1.99	1.32	2.28	1.21	0.21	1.93	-0.53	1.60	-0.74	-0.16	0.24	55.12
16	ALG	A	Good	45.9	0.37	0.70	0.19	-1.00	-1.99	-1.66	-2.07	1.42	-0.08	1.24	-0.37	0.14	66.33
17	BMB	B	Bad	51.1	0.28	0.75	1.55	0.17	-0.54	-1.17	-0.85	0.76	0.38	0.32	-1.12	0.19	59.96
18	AMG	A	Good	51.1	0.06	0.65	0.14	-1.08	-0.46	-1.11	-0.71	0.41	-0.25	0.53	-0.09	-0.29	62.74
19	CHB	C	Bad	55.9	0.02	-0.03	1.63	1.87	1.31	0.16	1.70	-0.55	0.75	-0.96	0.16	-0.36	55.36
20	BHB	B	Bad	55.9	-0.49	0.97	1.68	0.00	1.30	0.22	1.70	-0.91	0.50	-1.31	-1.23	-0.49	59.70
21	ALB	A	Bad	45.9	-0.59	-1.73	-0.53	-2.07	-2.08	-1.47	-1.95	1.25	-1.70	0.90	-0.51	-0.79	64.55
22	AHG	A	Good	55.9	-0.99	0.91	-0.10	-1.10	1.28	0.95	2.13	-1.49	-0.33	-1.30	-0.07	-1.11	64.13
23	AMB	A	Bad	51.1	-1.27	-1.57	-0.63	-2.21	-0.55	-0.73	-0.78	-0.11	-1.49	0.13	-0.42	-1.38	63.27
24	AHB	A	Bad	55.9	-1.85	-1.52	-0.21	-1.97	1.40	0.98	2.30	-1.61	-1.72	-1.65	0.01	-1.93	58.28

OP: Overall preference, AS: Acoustic satisfaction, VS: Visual satisfaction, SP: Speech privacy, VP: Visual privacy, WS: Work satisfaction, TP: Task performance.

References

- [1] V. Kupritz, Accommodating privacy to facilitate new ways of working, *J. Architect. Plann. Res.* 20 (2003) 122–135. <https://www.jstor.org/stable/43030651>.
- [2] T.J. Van Der Voordt, Productivity and employee satisfaction in flexible workplaces, *J. Corp. Real Estate* 6 (2004) 133–148. <https://doi.org/10.1108/14630010410812306>.
- [3] A. Kaarlela-Tuomaala, R. Helenius, E. Keskinen, V. Hongisto, Effects of acoustic environment on work in private office rooms and open-plan offices—longitudinal study during relocation, *Ergonomics* 52 (2009) 1423–1444. <https://doi.org/10.1080/00140130903154579>.
- [4] T.L. Smith-Jackson, K.W. Klein, Open-plan offices: task performance and mental workload, *J. Environ. Psychol.* 29 (2009) 279–289. <https://doi.org/10.1016/j.jenvp.2008.09.002>.
- [5] M. Zhang, J. Kang, F. Jiao, A social survey on the noise impact in open-plan working environments in China, *Sci. Total Environ.* 438 (2012) 517–526. <https://doi.org/10.1016/j.scitotenv.2012.08.082>.
- [6] J. Kim, R. De Dear, Workspace satisfaction: the privacy-communication trade-off in open-plan offices, *J. Environ. Psychol.* 36 (2013) 18–26. <https://doi.org/10.1016/j.jenvp.2013.06.007>.
- [7] Y. Al Horr, M. Arif, A. Kaushik, A. Mazroei, M. Katafygiotou, E. Elsarrag, Occupant productivity and office indoor environment quality: a review of the literature, *Build. Environ.* 105 (2016) 369–389. <https://doi.org/10.1016/j.buildenv.2016.06.001>.
- [8] Z. Li, Q. Zhang, F. Fan, S. Shen, A comprehensive comfort assessment method for indoor environmental quality in university open-plan offices in severe cold regions, *Build. Environ.* 197 (2021), 107845. <https://doi.org/10.1016/j.buildenv.2021.107845>.
- [9] S. Kang, D. Ou, C.M. Mak, The impact of indoor environmental quality on work productivity in university open-plan research offices, *Build. Environ.* 124 (2017) 78–89. <https://doi.org/10.1016/j.buildenv.2017.07.003>.
- [10] C.B. Danielsson, L. Bodin, Difference in satisfaction with office environment among employees in different office types, *J. Architect. Plann. Res.* 26 (2009) 241–257. <https://www.jstor.org/stable/43030872>.
- [11] A. Shafaghath, A. Keyvanfar, M.S. Ferwati, T. Alizadeh, Enhancing staff's satisfaction with comfort toward productivity by sustainable open-plan office design, *Sustain. Cities Soc.* 19 (2015) 151–164. <https://doi.org/10.1016/j.scs.2015.08.001>.
- [12] P.J. Lee, B.K. Lee, J.Y. Jeon, M. Zhang, J. Kang, Impact of noise on self-rated job satisfaction and health in open-plan offices: a structural equation modelling approach, *Ergonomics* 59 (2016) 222–234. <https://doi.org/10.1080/00140139.2015.1066877>.
- [13] S.P. Banbury, D.C. Berry, Office noise and employee concentration: identifying causes of disruption and potential improvements, *Ergonomics* 48 (2005) 25–37. <https://doi.org/10.1080/00140130412331311390>.
- [14] M. Pierrette, E. Parizet, P. Chevret, Perception and evaluation of noise sources in open-plan office, in: *Proceedings of Meetings on Acoustics ICA2013*, vol. 19, J. Acoust. Soc. Am., 2013, 040127. <https://doi.org/10.1121/1.4800003>.
- [15] E. Sundstrom, J.P. Town, R.W. Rice, D.P. Osborn, M. Brill, Office noise, satisfaction, and performance, *Environ. Behav.* 26 (1994) 195–222. <https://doi.org/10.1177/001391659402600204>.
- [16] H. Jahncke, S. Hygge, N. Halin, A.M. Green, K. Dimberg, Open-plan office noise: cognitive performance and restoration, *J. Environ. Psychol.* 31 (2011) 373–382. <https://doi.org/10.1016/j.jenvp.2011.07.002>.
- [17] S. Banbury, D.C. Berry, Disruption of office-related tasks by speech and office noise, *Br. J. Psychol.* 89 (1998) 499–517. <https://doi.org/10.1111/j.2044-8295.1998.tb02699.x>.
- [18] M. Haka, A. Haapakangas, J. Keränen, J. Hakala, E. Keskinen, V. Hongisto, Performance effects and subjective disturbance of speech in acoustically different office types—a laboratory experiment, *Indoor Air* 19 (2009) 454–467. <https://doi.org/10.1111/j.1600-0668.2009.00608.x>.
- [19] M. Pierrette, E. Parizet, P. Chevret, J. Chatillon, Noise effect on comfort in open-space offices: development of an assessment questionnaire, *Ergonomics* 58 (2015) 96–106. <https://doi.org/10.1080/00140139.2014.961972>.
- [20] K.P. Waye, R. Rylander, S. Benton, H.G. Leventhall, Effects on performance and work quality due to low frequency ventilation noise, *J. Sound Vib.* 20 (1997) 467–474. <https://doi.org/10.1006/jsvi.1997.1013>.
- [21] P. Roelofs, Performance loss in open-plan offices due to noise by speech, *J. Facil. Manag.* 6 (2008) 202–211. <https://doi.org/10.1108/14725960810885970>.
- [22] E. Sundstrom, R.K. Herbert, D.W. Brown, Privacy and communication in an open-plan office: a case study, *Environ. Behav.* 14 (1982) 379–392. <https://doi.org/10.1177/0013916582143007>.
- [23] K. Duvall-Early, J.O. Benedict, The relationships between privacy and different components of job satisfaction, *Environ. Behav.* 24 (1992) 670–679. <https://doi.org/10.1177/0013916592245006>.
- [24] V. Hongisto, A. Haapakangas, J. Varjo, R. Helenius, H. Koskela, Refurbishment of an open-plan office—environmental and job satisfaction, *J. Environ. Psychol.* 45 (2016) 176–191. <https://doi.org/10.1016/j.jenvp.2015.12.004>.
- [25] J. Goins, J. Jellema, H. Zhang, Architectural enclosure's effect on office worker performance: a comparison of the physical and symbolic attributes of workspace dividers, *Build. Environ.* 45 (2010) 944–948. <https://doi.org/10.1016/j.buildenv.2009.09.015>.
- [26] M.J. O'Neill, P. Carayon, The relationship between privacy, control, and stress responses in office workers, in: *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, vol. 37, SAGE Publications, Sage CA: Los Angeles, CA, 1993, pp. 479–483. <https://doi.org/10.1177/154193129303700702>.
- [27] A.L. Brown, A review of progress in soundscapes and an approach to soundscape planning, *Int. J. Acoust. Vib.* 17 (2012) 73–81.
- [28] J. Kang, B. Schulte-Fortkamp, *Soundscape and the Built Environment*, CRC Press, Boca Raton, FL, 2018.
- [29] ISO 12913-1, *Acoustics—Soundscape—Part 1: Definition and Conceptual Framework*, International Organization for Standardization, Geneva, Switzerland, 2014.
- [30] J.Y. Jeon, P.J. Lee, J.Y. Hong, D. Cabrera, Non-auditory factors affecting urban soundscape evaluation, *J. Acoust. Soc. Am.* 130 (2011) 3761–3770. <https://doi.org/10.1121/1.3652902>.

- [31] M. Lionello, F. Aletta, J. Kang, A systematic review of prediction models for the experience of urban soundscapes, *Appl. Acoust.* 170 (2020), 107479, <https://doi.org/10.1016/j.apacoust.2020.107479>.
- [32] V. Acun, S. Yilmazer, Combining grounded theory (GT) and structural equation modelling (SEM) to analyze indoor soundscape in historical spaces, *Appl. Acoust.* 155 (2019) 515–524, <https://doi.org/10.1016/j.apacoust.2019.06.017>.
- [33] S. Torresin, F. Aletta, F. Babich, E. Bourdeau, J. Harvie-Clark, J. Kang, L. Lavia, A. Radicchi, R. Albatici, Acoustics for supportive and healthy buildings: emerging themes on indoor soundscape research, *Sustainability* 12 (2020) 6054, <https://doi.org/10.3390/su12156054>.
- [34] P.N. Dokmeci Yorukoglu, J. Kang, Development and testing of indoor soundscape questionnaire for evaluating contextual experience in public spaces, *Build. Acoust.* 24 (2017) 307–324, <https://doi.org/10.1177/1351010X17743642>.
- [35] S. Torresin, R. Albatici, F. Aletta, F. Babich, T. Oberman, S. Siboni, J. Kang, Indoor soundscape assessment: a principal components model of acoustic perception in residential buildings, *Build. Environ.* 182 (2020), 107152, <https://doi.org/10.1016/j.buildenv.2020.107152>.
- [36] P.N. Dokmeci Yorukoglu, J. Kang, Analysing sound environment and architectural characteristics of libraries through indoor soundscape framework, *Arch. Acoust. Q.* 41 (2016) 203–212.
- [37] U.B. Ercakmak, P.N. Dokmeci Yorukoglu, Comparing Turkish and European noise management and soundscape policies: a proposal of indoor soundscape integration to architectural design and application, *Acoustics* 1 (2019) 847–865, <https://doi.org/10.3390/acoustics1040051>.
- [38] J. Carles, F. Bernáldez, J.D. Lucio, Audio-visual interactions and soundscape preferences, *Landscape Res.* 17 (1992) 52–56, <https://doi.org/10.1080/01426399208706361>.
- [39] J.Y. Hong, J.Y. Jeon, Designing sound and visual components for enhancement of urban soundscapes, *J. Acoust. Soc. Am.* 134 (2013) 2026–2036, <https://doi.org/10.1121/1.4817924>.
- [40] H. Li, S.K. Lau, A review of audio-visual interaction on soundscape assessment in urban built environments, *Appl. Acoust.* 166 (2020), 107372, <https://doi.org/10.1016/j.apacoust.2020.107372>.
- [41] J.Y. Jeon, H.I. Jo, Effects of audio-visual interactions on soundscape and landscape perception and their influence on satisfaction with the urban environment, *Build. Environ.* 169 (2020), 106544, <https://doi.org/10.1016/j.buildenv.2019.106544>.
- [42] S. Torresin, G. Pernigotto, F. Cappelletti, A. Gasparella, Combined effects of environmental factors on human perception and objective performance: a review of experimental laboratory works, *Indoor Air* 28 (2018) 525–538, <https://doi.org/10.1111/ina.12457>.
- [43] M. Schweiker, E. Ampatzi, M.S. Andargie, R.K. Andersen, E. Azar, V.M. Barthelmes, C. Berger, L. Bourikas, S. Carlucci, G. Chinazzo, L. Edappilly, Review of multi-domain approaches to indoor environmental perception and behaviour, *Build. Environ.* 176 (2020), 106804, <https://doi.org/10.1016/j.buildenv.2020.106804>.
- [44] H.I. Jo, J.Y. Jeon, The influence of human behavioral characteristics on soundscape perception in urban parks: subjective and observational approaches, *Landsc. Urban Plann.* 203 (2020), 103890, <https://doi.org/10.1016/j.landurbplan.2020.103890>.
- [45] Y. Hasegawa, S.K. Lau, Audiovisual bimodal and interactive effects for soundscape design of the indoor environments: a systematic review, *Sustainability* 13 (2021) 339, <https://doi.org/10.3390/su13010339>.
- [46] H.I. Jo, J.Y. Jeon, Downstairs resident classification characteristics for upstairs walking vibration noise in an apartment building under virtual reality environment, *Build. Environ.* 150 (2019) 21–32, <https://doi.org/10.1016/j.buildenv.2018.12.054>.
- [47] J.Y. Jeon, H.I. Jo, Three-dimensional virtual reality-based subjective evaluation of road traffic noise heard in urban high-rise residential buildings, *Build. Environ.* 148 (2019) 468–477, <https://doi.org/10.1016/j.buildenv.2018.11.004>.
- [48] Z. Abdalrahman, L. Galbrun, Audio-visual preferences, perception, and use of water features in open-plan offices, *J. Acoust. Soc. Am.* 147 (2020) 1661–1672, <https://doi.org/10.1121/1.0000892>.
- [49] M. Navai, J.A. Veitch, *Acoustic Satisfaction in Open-Plan Offices: Review and Recommendations*, Institute for Research in Construction National Research Council, Ottawa, Canada, 2003.
- [50] I. Balazova, G. Clausen, J.H. Rindel, T. Poulsen, D.P. Wyon, Open-plan office environments: a laboratory experiment to examine the effect of office noise and temperature on human perception, comfort and office work performance, *Indoor Air* (2008) 17–22.
- [51] P.J. Lee, J.Y. Jeon, A laboratory study for assessing speech privacy in a simulated open-plan office, *Indoor Air* 24 (2014) 307–314, <https://doi.org/10.1111/ina.12083>.
- [52] J.Y. Jeon, H.I. Jo, S.M. Kim, H.S. Yang, Subjective and objective evaluation of water-supply and drainage noises in apartment buildings by using a head-mounted display, *Appl. Acoust.* 148 (2019) 289–299, <https://doi.org/10.1016/j.apacoust.2018.12.037>.
- [53] T. Hong, M. Lee, S. Yeom, K. Jeong, Occupant responses on satisfaction with window size in physical and virtual built environments, *Build. Environ.* 166 (2019), 106409, <https://doi.org/10.1016/j.buildenv.2019.106409>.
- [54] J.Y. Hong, B. Lam, Z.T. Ong, K. Ooi, W.S. Gan, J. Kang, J. Feng, S.T. Tan, Quality assessment of acoustic environment reproduction methods for cinematic virtual reality in soundscape applications, *Build. Environ.* 149 (2019) 1–14, <https://doi.org/10.1016/j.buildenv.2018.12.004>.
- [55] C.J. Wilson, A. Soranzo, The use of virtual reality in psychology: a case study in visual perception, *Comput. Math Methods Med.* 2015 (2015), 151702, <https://doi.org/10.1155/2015/151702>.
- [56] G. Charness, U. Gneezy, M.A. Kuhn, Experimental methods: between-subject and within-subject design, *J. Econ. Behav. Organ.* 81 (2012) 1–8, <https://doi.org/10.1016/j.jebo.2011.08.009>.
- [57] ISO 3382-3:2012, *Acoustics — Measurement of Room Acoustic Parameters — Part 3: Open-Plan Offices*, International Organization for Standardization, Geneva, Switzerland, 2012.
- [58] ISO 3382-1:2009, *Acoustics — Measurement of Room Acoustic Parameters — Part 1: Performance Spaces*, International Organization for Standardization, Geneva, Switzerland, 2009.
- [59] B.K. Lee, *Acoustic Evaluation and Design of Open-Plan Offices through Survey and Computer Simulation*, Graduate School of Hanyang University, 2016.
- [60] P. Virjonen, J. Keränen, V. Hongisto, Determination of acoustical conditions in open-plan offices: proposal for new measurement method and target values, *Acta Acust. United Ac.* 95 (2009) 279–290, <https://doi.org/10.3813/AAA.918150>.
- [61] A. Haapakangas, V. Hongisto, M. Eerola, T. Kuusisto, Distraction distance and perceived disturbance by noise—an analysis of 21 open-plan offices, *J. Acoust. Soc. Am.* 141 (2017) 127–136, <https://doi.org/10.1121/1.4973690>.
- [62] M. Yadav, D. Cabrera, J. Kim, J. Fels, R. de Dear, Sound in occupied open-plan offices: objective metrics with a review of historical perspectives, *Appl. Acoust.* 177 (2021), 107943, <https://doi.org/10.1016/j.apacoust.2021.107943>.
- [63] S.H. Park, P.J. Lee, B.K. Lee, M. Roskams, B.P. Haynes, Associations between job satisfaction, job characteristics, and acoustic environment in open-plan offices, *Appl. Acoust.* 168 (2020), 107425, <https://doi.org/10.1016/j.apacoust.2020.107425>.
- [64] M. Roskams, B. Haynes, P.J. Lee, S.H. Park, Acoustic comfort in open-plan offices: the role of employee characteristics, *J. Corp. Real Estate.* 21 (2019) 254–270, <https://doi.org/10.1108/JCRE-02-2019-0011>.
- [65] S. Delle Macchie, S. Secchi, G. Cellai, Acoustic issues in open plan offices: a typological analysis, *Buildings* 8 (2018) 161, <https://doi.org/10.3390/buildings8110161>.
- [66] C.R.M. Passero, P.H.T. Zannin, Acoustic evaluation and adjustment of an open-plan office through architectural design and noise control, *Appl. Ergon.* 43 (2012) 1066–1071, <https://doi.org/10.1016/j.apergo.2012.03.007>.
- [67] ISO 22955:2021, *Acoustics—Acoustic Quality of Open Office Spaces*, ISO, Geneva, Switzerland, 2021.
- [68] H. Ma, S. Shu, An experimental study: the restorative effect of soundscape elements in a simulated open-plan office, *Acta Acust. United Ac.* 104 (2018) 106–115, <https://doi.org/10.3813/AAA.919150>.
- [69] L. Brocolini, E. Parizet, P. Chevret, Effect of masking noise on cognitive performance and annoyance in open plan offices, *Appl. Acoust.* 114 (2016) 44–55, <https://doi.org/10.1016/j.apacoust.2016.07.012>.
- [70] D.L. Woods, M.M. Kishiyama, E.W. Yund, T.J. Herron, B. Edwards, O. Poliva, R. F. Hink, B. Reed, Improving digit span assessment of short-term verbal memory, *J. Clin. Exp. Neuropsychol.* 33 (2011) 101–111, <https://doi.org/10.1080/13803395.2010.493149>.
- [71] C. Lavandier, B. Defréville, The contribution of sound source characteristics in the assessment of urban soundscapes, *Acta Acust. United Ac.* 92 (2006) 912–921.
- [72] J.Y. Jeon, H.I. Jo, R. Seo, K.H. Kwak, Objective and subjective assessment of sound diffuseness in musical venues via computer simulations and a scale model, *Build. Environ.* 173 (2020), 106740, <https://doi.org/10.1016/j.buildenv.2020.106740>.
- [73] S. Yeom, H. Kim, T. Hong, M. Lee, Determining the optimal window size of office buildings considering the workers' task performance and the building's energy consumption, *Build. Environ.* 177 (2020), 106872, <https://doi.org/10.1016/j.buildenv.2020.106872>.
- [74] K. Yildirim, A. Akalin-Baskaya, M. Celebi, The effects of window proximity, partition height, and gender on perceptions of open-plan offices, *J. Environ. Psychol.* 27 (2007) 154–165, <https://doi.org/10.1016/j.jenvp.2007.01.004>.
- [75] S. Torresin, R. Albatici, F. Aletta, F. Babich, T. Oberman, A.E. Stawinoga, J. Kang, Indoor soundscapes at home during the COVID-19 lockdown in London—Part I: associations between the perception of the acoustic environment, occupants' activity and well-being, *Appl. Acoust.* 183 (2021), 108305, <https://doi.org/10.1016/j.apacoust.2021.108305>.
- [76] G. Halkos, D. Bousinakis, The effect of stress and satisfaction on productivity, *Int. J. Prod. Perform. Manag.* 59 (2010) 415–431, <https://doi.org/10.1108/17410401011052869>.
- [77] C. Wang, F. Zhang, J. Wang, J.K. Doyle, P.A. Hancock, C.M. Mak, S. Liu, How indoor environmental quality affects occupants' cognitive functions: a systematic review, *Build. Environ.* 193 (2021), 107647, <https://doi.org/10.1016/j.buildenv.2021.107647>.
- [78] E. Parizet, P. Chevret, K. Kostallari, A simple sound metric for evaluating sound annoyance in open-plan offices, *J. Acoust. Soc. Am.* 141 (2017), 3966–3966.
- [79] R.J. Baken, R.F. Orlikoff, *Clinical Measurement of Speech and Voice*, Cengage Learning, Boston, MA, 2020.
- [80] D.H. Brush, M.K. Moch, A. Pooyan, Individual demographic differences and job satisfaction, *J. Organ. Behav.* 8 (1987) 139–155, <https://doi.org/10.1002/job.4030080205>.